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Testing for Granger causality in heterogeneous panels with cross-sectional dependence

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Abstract

This paper proposes a panel Granger causality approach for heterogeneous panels with cross-sectional dependence. We define a panel VAR model with unobserved common factors and apply the PANIC procedure to obtain the de-factored data. We then estimate the lag augmented (LA)-VAR model for each cross section and construct the panel statistics based on the meta-analytic approach that combines the *p*-values of the individual statistics. The Monte Carlo simulations indicate that the combination tests show good size and power properties and appear suitable for the panels where cross sections may have different unit root or co-integration properties. We finally re-investigate Granger causality between export and economic growth in OECD countries. The results shed light on the importance of accounting for cross-sectional dependence within a factor model framework in determining direction of Granger causality for country-specific analysis. The results further reveal that export and economic growth do not cause each other in the majority of the European Union countries.

Keywords Granger causality · Panel data · Common factor

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1 Introduction

It is well documented that panel data methods offer more powerful unit root and co-integration tests. The literature indicates the importance of considering cross-sectional heterogeneity and cross-sectional dependence within a panel framework. In parallel to panel unit root and co-integration methods,[1] testing for panel Granger causality has attracted interest and different approaches have recently been developed.

One attempt to test Granger causality in panel data is based on estimating a panel vector autoregressive or vector error correction model by means of the generalized method of moments (GMM)[2] (see Holtz-Eakin et al. 1988), which does not consider neither heterogeneity nor cross-sectional dependence. The panel Granger causality method proposed by Kónya (2006) accounts for cross-sectional dependence and heterogeneity by employing the seemingly unrelated regressions (SUR) of Zellner (1962). As it is well known, the SUR estimator is only feasible when time dimension (T) is reasonably larger than cross-section dimension (N). The panel Granger causality methods proposed by Dumitrescu and Hurlin (2012) and Emirmahmutoglu and Kose (2011) focus on considering cross-sectional heterogeneity by extending Granger (1969) and Toda and Yamamoto (1995) approaches into the panel framework, which work best when N is smaller than T. In a more recent study, Juodis et al. (2021) propose a novel approach to testing for Granger causality in heterogeneous panels, which can be used in large N cases and works best for cases where N is same to or larger than T.

Early contributions in the literature on panel unit root and co-integration tests benefit from bootstrapping to account for cross-sectional dependence.[3] Emirmahmutoglu and Kose (2011) and Dumitrescu and Hurlin (2012) similarly employ bootstrap distributions for the panel statistics. As discussed in Hadri et al. (2015), bootstrapping accommodates cross-sectional dependence of general form and does not provide information on the structure of dependency. Factor models have gained their importance to account for cross-correlations in panel data during the last decades.[4] The intuition behind using factor models is that the global shocks or common factors can affect each cross section with different weights and intensities, depending on their intrinsic properties (Karabiyik et al. 2019).

This paper focuses on testing for Granger causality in heterogeneous panels by accounting for cross-sectional dependence. We extend Emirmahmutoglu and Kose

[1] We refer an interested reader to Choi (2006, 2015) and Breitung and Das (2008) for the comprehensive reviews on the panel unit root and co-integration methods.

[2] The testing framework based on the GMM is appealing when time dimension (T) of the panel is short and coefficients are homogeneous. Pesaran and Smith (1995) show that the traditional approaches such as the fixed effects, instrumental variables, and the GMM can yield inconsistent and potentially misleading estimates of the parameters in dynamic panel models with large T unless the coefficients are identical.

[3] See, among others, Maddala and Wu (1999), Smith et al. (2004), and Westerlund (2007).

[4] See, among others, Bai and Ng (2004), Phillips and Sul (2003), Moon and Perron (2004), Pesaran (2006, 2007), and Westerlund (2008).

(2011) approach within the context of factor models. Specifically, we define a panel VAR model with unobserved common factors and apply the PANIC approach of Bai and Ng (2004) to consistently estimate common factors for obtaining the de-factored data. We then estimate the lag augmented (LA)-VAR model for each cross section to obtain the individual Wald statistics and construct the panel statistics based on the meta-analytic approach by combining the *p*-values of individual Wald statistics.

To examine the small sample performance of proposed tests, we consider different data-generating processes with different unit root and co-integration properties for cross-sectional units. The Monte Carlo simulations indicate that (i) the tests show mild size distortions when N is not smaller than T, which disappear as T grows and N is fixed; (ii) they have good empirical power even in small samples; and (iii) our approach suits the dependent panels where cross-sections may have different unit root or co-integration properties. The simulations also reveal that the proposed tests show better size and power properties than the bootstrap tests of Emirmahmutoglu and Kose (2011) and Dumitrescu and Hurlin (2012) as T grows.

We finally re-investigate Granger causality between export and economic growth for OECD countries within a comprehensive framework by conducting existing and new testing procedures. While the panel results support two-way Granger causality, the country-specific results indicate the importance of accounting for cross-sectional dependence. The approach proposed in this study also unveils less evidence on Granger causality between export and economic growth in in the majority of the European Union (EU) countries.

The rest of the paper is as follows. The next section describes the model and testing procedures. Section 3 is devoted to small sample analysis. Section 4 illustrates an empirical application of the proposed approach compared to the existing methods. The concluding remarks are finally presented in Sect. 5.

2 Model and procedures

We consider panel VAR model with m-dimensional series $Z_{it}^{cd} = \left(Z_{1,it}^{cd}, \ldots, Z_{m,it}^{cd}\right)', i = 1, \ldots, N$ and $t = 1, \ldots, T$, given by[5]

$$Z_{it}^{cd} = \mu_i + \Lambda_i{}' F_t + Z_{it}, \tag{1}$$

$$F_t = C(L)u_t, \tag{2}$$

$$Z_{it} = A_{i,1} Z_{i,t-1} + \ldots + A_{i,p_i} Z_{i,t-p_i} + \varepsilon_{it} \tag{3}$$

[5] A similar data generating process is already considered by Carrion-i-Silvestre and Surdeanu (2011), Örsal and Arsova (2017) and Arsova and Örsal (2018) for testing panel co-integration. Our setting is very comparable to those of Carrion-i-Silvestre and Surdeanu (2011) and Arsova and Örsal (2018), and only difference is the research question for testing Granger causality instead of co-integration. We are also in line with Carrion-i-Silvestre and Surdeanu (2011) and Örsal and Arsova (2017) by constructing panel statistic based on the combination of individual *p*-values.

where μ_i is unknown m-dimensional fixed effects, $F_t = (F_{1t}, \ldots, F_{rt})'$ is r-dimensional vector of unobservable common factors that r is same for all i, Λ_i are $(r \times m)$ matrix of factor loadings, Z_{it} are the idiosyncratic components, $A_{i1}, \ldots, A_{i,p_i}$ are $m \times m$ matrices of coefficients, and $p_i = k_i + d_i$ where k_i is the lag order of VAR model and d_i is the maximum integration order of Z_{it}.

Assumption 1 The common factor F_t is a covariance stationary process with $u_t \sim iid(0, \Sigma_u)$, $\mathrm{E}\,\|u_t\|^4 \leq M < \infty$, $\sum_{j=0}^{\infty} j\|C_j\| < M < \infty$, where $C(L)$ is an $(r \times r)$ matrix consisting of polynomials of the lag operator L that $C(L) = \sum_{j=0}^{\infty} C_j L^j$.[6]

Assumption 2 For factor loadings, Λ_i is deterministic and $\|\Lambda_i\| \leq M$, or Λ_i is stochastic and $\mathrm{E}\,\|\Lambda_i\|^4 \leq M < \infty$; and $N^{-1}\sum_{i=1}^{N} \Lambda_i \Lambda_i' \to \Sigma_\Lambda$ as $N \to \infty$ where Σ_Λ is a non-random positive definite matrix. Thereby, enough loadings must be non-zero; and are bounded away from zero and above both in finite samples and in asymptotic. This ensures de-factoring the data, and the factor model is hence identified.[7]

Assumption 3 The idiosyncratic components Z_{it} are either I(1) or I(0).[8]

Assumption 4 The idiosyncratic errors ε_{it} are independently and identically distributed with $E(\varepsilon_{it}) = 0$, $E\left(\varepsilon_{it}^2\right) = \Sigma_{\varepsilon_i} 0$ which means that Σ_{ε_i} is positive definite and $E\left(\varepsilon_{it}\varepsilon_{js}\right) = 0\ \forall i \neq j$ and $\forall(t, s)$.

Assumption 5 Λ_i, u_t and ε_{it} are mutually independently distributed across i and t

Assumption 6 The lag order k_i is known and may differ across cross-sections.[9]

2.1 De-factorization procedure

For testing Granger causality in dependent panels, we first need to remove common factors from the data. To estimate unobserved common factors, two methods, known as the PANIC approach of Bai and Ng (2004) and the PANICCA -PANIC on CA (cross-sectional averages)[10]- approach of Reese and Westerlund (2016) are the workhorses of the industry. The idea behind both approaches is to take the first difference of the data as $\Delta Z_{it}^{cd} = \Lambda_i{}'\Delta F_t + \Delta Z_{it}$, which ensures the stationarity and avoids the

[6] Even though we assume stationarity of F_t, it is not strictly necessary since we de-factor the data using PANIC procedure which is robust to non-stationary factors. Nonetheless, following Westerlund and Edgerton (2008), by assuming stationarity of F_t we ensure that the order of integration of Z_{it}^{cd} depends only on the integration degree of the idiosyncratic components Z_{it}.

[7] The assumptions for the factor innovations and loadings are the standard in the factor model literature.

[8] An exposition about the order of integration for VAR system with integrated variables can be found in Lütkepohl (1991, Chapter 11). Accordingly, Z_{it} is $I(0)$ if the roots of polynomial $|A_i(s)| = 0$ lie outside the unit circle and Z_{it} is $I(1)$ if one root equals to one and all other roots outside the unit circle, where $A_i(s) = I_m - A_{i1}s - \cdots - A_{ip_i}s^{p_i}$ for each $i = 1, \ldots, N$. The order of integration of Z_{it} is $I(1)$ if there is just one $I(1)$ component in Z_{it} and all other components are $I(0)$.

[9] In practice, the lag order is rarely known and can be determined by Akaike or Schwarz information criterion.

[10] The CA approach is originally proposed by Pesaran (2006) and it then is used in either panel unit root (Pesaran 2007) or panel co-integration framework (Banerjee and Carrion-i-Silvestre 2015).

uncertainty regarding the integration order of Z_{it}^{cd}. The common and idiosyncratic components can thereby be estimated using existing approaches for common factor models (Reese and Westerlund 2016). The PANIC approach is based on estimating F_t by principal components analysis (PCA) and the PANICCA approach uses the cross-sectional averages of observed data as the estimated factors. The estimated $\widehat{F}_t$, obtained by accumulating $\Delta\widehat{F}_t$, is consistent for (the space spanned by) F_t (see Bai and Ng 2004; Reese and Westerlund 2016), and extracting the estimated common components from the observed data yields the idiosyncratic components $\widehat{Z}_{it} = \sum_{s=2}^{t}(\Delta Z_{is}^{cd} - \widehat{\Lambda}_i'\Delta\widehat{F}_s)$. This procedure gives $\widehat{Z}_{it}$ to be consistent for Z_{it} under very general conditions and employing $\widehat{Z}_{it}$ in testing procedure is equivalent to using Z_{it} (see Juodis and Westerlund 2019, p. 68). We thereby can test for Granger non-causality hypothesis with the usual Wald test by using estimated idiosyncratic components $\widehat{Z}_{it}$.

Remark 1: The PANIC approach consistently estimates the space spanned by true factors as $N, T \to \infty$ simultaneously (see Bai and Ng 2004). The PANICCA has the same asymptotic theory as the PANIC, with improved small sample performance especially for the panels with small to medium-N (see Reese and Westerlund 2016). Westerlund and Urbain (2015) indicate that both approaches are asymptotical equivalent if $N/T \to 0$, but are biased when $N/T \to \tau > 0$.

Remark 2: In the PANIC approach, we need to estimate the number of factors *(r)*, which can be consistently estimated by assuming r to be same for all cross sections using the following criterion function proposed by Bai and Ng (2002)

$$IC(\hat{r}) = \ln\left(V\left(\hat{r}, \hat{F}^{\hat{r}}\right)\right) + \hat{r} \cdot \mathrm{g} \tag{4}$$

where $V\left(\widehat{r}, \widehat{F^r}\right)$ is obtained by using estimated factors and factor loadings in $V\left(\widehat{r}, \widehat{F^r}\right) = \min_{\lambda} \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=2}^{T}\left(z_{it}^{cd} - \widehat{\Lambda}_i^{\widehat{r}}\widehat{f}_t^{\widehat{r}}\right)^2$ where $z_{it}^{cd} = \Delta Z_{it}^{cd}$ and $\widehat{f}_t^{\widehat{r}} = \Delta\widehat{F}_t^{\widehat{r}}$. Then $\widehat{r}$, the consistent estimate of r, is selected by minimizing $IC(\widehat{r})$. The penalty function (g) in Bai and Ng (2002) depends on N and T, given by $\mathrm{g} = (N+T)NT^{-1}\ln C_{NT}^2$ for $IC_2(\widehat{r})$ criterion with $\mathrm{C_{NT}} = \min\left(\sqrt{N}, \sqrt{T}\right)$. Nonetheless, for the panel VAR framework, the minimization problem requires defining $V\left(\widehat{r}, \widehat{F^r}\right)$ as $V\left(\widehat{r}, \widehat{F^r}\right) = \min_{\lambda} \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=2}^{T}\sum_{q=1}^{m}\left(z_{it,q}^{cd} - \widehat{\Lambda}_{i,q}^{\widehat{r}}\widehat{f}_t^{\widehat{r}}\right)^2$. Hence, the penalty function also depends on the number of endogenous variables (m), given by $\mathrm{C_{mNT}} = \min\left(\sqrt{mN}, \sqrt{T}\right)$. Note that even though the effect of m in the criterion function is negligible as N and T go to infinity, it may matter in small samples [see Theorem 2.1 in Huang (2008, p. 223)]. After determining r, the factor loadings are obtained by $\sqrt{mN}$ times eigenvalues corresponding to the r largest eigenvectors of $z'z$. Given $\widehat{\Lambda}$, factors are estimated by $\widehat{f} = z\widehat{\Lambda}/(mN)$.

2.2 Granger causality procedure

As indicated in Sims et al. (1990), the Wald statistic for Granger non-causality hypothesis based on VAR estimation with level variables has a non-standard asymptotic

distribution and depends on nuisance parameters if VAR process has unit root or co-integration. To overcome this issue, Toda and Yamamoto (1995) suggest an intuitive and simple solution based on estimating LA-VAR model by using level variables, which is robust to unit root and co-integration properties of variables and makes the conventional asymptotic theory applicable. More precisely, the LA-VAR is based on overfitting VAR*(k)* model with d extra lags (where d is the maximum order of integration) and imposing the restrictions for hypothesis testing on the k coefficients.

To estimate the panel LA-VAR, by assuming $m = 1$ without loss of generality let us partition Z_{it} as $(y_{it}, x_{it})'$ and define a bivariate form of Eq. (3) as

$$y_{it} = \sum_{j=1}^{k_i} A_{11,ij} y_{i,t-j} + \sum_{j=1}^{k_i} A_{12,ij} x_{i,t-j} + \sum_{j=k_i+1}^{k_i+d_i} A_{11,ij} y_{i,t-j} + \sum_{j=k_i+1}^{k_i+d_i} A_{12,ij} x_{i,t-j} + \varepsilon_{1it} \tag{5a}$$

$$x_{it} = \sum_{j=1}^{k_i} A_{21,ij} y_{i,t-j} + \sum_{j=1}^{k_i} A_{22,ij} x_{i,t-j} + \sum_{j=k_i+1}^{k_i+d_i} A_{21,ij} y_{i,t-j} + \sum_{j=k_i+1}^{k_i+d_i} A_{22,ij} x_{i,t-j} + \varepsilon_{2it}. \tag{5b}$$

The null hypothesis of Granger non-causality from x_{it} to y_{it} is formulated as a set of linear restrictions on $A_{12,i}$ given by.

$H_0 : A_{12,i} = 0 \forall i = 1, \ldots, N$ with $A_{12,i} = (A_{12,i1}, \ldots, A_{12,ik_i})'$.

The alternative hypothesis of Granger causality is defined as

$$H_1 : A_{12,i} \neq 0 \forall i = 1, \ldots, N_1$$
$$A_{12,i} = 0 \forall i = N_1 + 1, \ldots, N.$$

Let $A_i = vec(A'_{i1}, \ldots, A'_{i,p_i})'$ with $p_i = k_i + d_i$ for $i = 1, .., N$ be the vector of all VAR coefficients. The null hypothesis can be written as $R_i A_i = 0$ where R_i is a $k_i \times 2(k_i + d_i)$ matrix with $R_i = \left[0_{k_i \times k_i + d_i}, I_{k_i}, 0_{k_i \times d_i}\right]$. The Wald statistic for testing $H_0 : R_i A_i = 0$ against $R_i A_i \neq 0$ for each i is obtained by

$$W_i = T_i \left(R_i \hat{A}_i\right)' \left(R_i \hat{\Sigma}_i R_i'\right)^{-1} \left(R_i \hat{A}_i\right) \tag{6}$$

where $\widehat{A_i}$ is the estimate of parameter A_i under the alternative hypothesis, $\widehat{\Sigma}_i$ is the estimate of the variance–covariance matrix of ε_i, and T_i is the number of observations used to estimate (5).

Conjecture 1 Let $\left\{Z_{it}^{cd}\right\}_{i=1,t=1}^{N,T}$, be a $(m \times 1)$ vector of stochastic process with DGP given by (1) to (3). Under the null hypothesis of $H_0 : R_i A_i = 0$, as $N, T \to \infty$ simultaneously with $N/T \to 0$, W_i has an asymptotic χ^2 distribution with k_i degrees of freedom and diverges to $+\infty$(i.e., consistent) under the alternative hypothesis of $R_i A_i \neq 0$.

This conjecture is supported by simulation evidence to verify no consequence of the de-factorization procedure on the asymptotic properties of W_i statistic.[11] We conduct Monte Carlo simulations to investigate how W_i applied to the estimated idiosyncratic component (denoted as $W_{\widehat{z}}$) behaves in comparison with it applied to the true idiosyncratic component (denoted as W_z). The simulation results show that the distributions of $W_{\widehat{z}}$ and W_z statistics appear identical and match the asymptotic χ^2 distribution well. Moreover, $W_{\widehat{z}}$ and W_z statistics almost have the same mean, variance, and rejection rate under the null hypothesis (see Appendix).

To test the null hypothesis of panel Granger non-causality, we construct the panel statistic by combining the significance levels of W_i as advocated by Maddala and Wu (1999). The combination statistic proposed by Fisher (1932) is defined as

$$P = -2\sum_{i=1}^{N} \ln(\pi_i) \tag{7}$$

where π_i is the p-value of W_i. While the Fisher-type statistic is suggested under the assumption of cross-sectional independence, Bai and Ng (2004, p. 1140) note that the independence assumption can be relaxed by averaging the p-values over many cross-sectional units as N increases, and the same argument is further utilized in Bai and Carrion-i-Silvestre (2009) and Nazlioglu et al. (2023), *among others*.[12] Bai and Ng (2004) show that univariate tests for the estimated idiosyncratic component do not depend on the factor structure, and thus if the true idiosyncratic component Z_{it} is independent across i, tests based on the estimated idiosyncratic component $\widehat{Z}_{it}$ are asymptotically independent across i. The p-values are thereby independent U[0,1] random variables, which implies $-2\ln(\pi_i)$ is a χ^2 random variable with two degrees of freedom. Accordingly, P has a χ^2_{2N} distribution under the null hypothesis for fixed N. Choi (2001) modifies P test to allow $N \to \infty$, as

$$P_m = \frac{-2\sum_{i=1}^{N} \ln(\pi_i) - 2N}{\sqrt{4N}}. \tag{8}$$

Conjecture 2 As $T \to \infty$ and followed $N \to \infty$ *or* as $T, N \to \infty$ simultaneously with $N/T \to 0$, P_m has a standard normal distribution under the null hypothesis. If $N_1/N \to c$ (a fixed constant) as $N \to \infty$, it diverges to $+\infty$ under the alternative hypothesis.

For a basis of Conjecture 2, the modified test statistic can be written as

$$P_m = \sum_{i=1}^{N}\left(\frac{-2\ln(\pi_i) - E[-2\ln(\pi_i)]}{\sqrt{var[-2\ln(\pi_i)]}}\right) = \frac{1}{\sqrt{N}}\sum_{i=1}^{N}(-\ln(\pi_i) - 1) \tag{8a}$$

where $E[-2\ln(\pi_i)] = 2$ and $var[-2\ln(\pi_i)] = 4$, and $P_m \sim N(0, 1)$ as $T \to \infty$ and then $N \to \infty$ under the null hypothesis (Choi 2001, p. 255). Phillips and Moon (1999)

[11] We are grateful to associate editor (C. Hanck) and an anonymous reviewer for suggesting to support the conjecture by means of simulation evidence.

[12] Note that these studies focus on panel unit root tests instead of panel Granger causality tests.

show that this result obtained under the sequential limits holds under joint limits when $T, N \to \infty$ simultaneously with the rate condition $N/T \to 0$. For the joint limits, let

$$\sum_{i=1}^{N} \ln(\pi_i) = \ln\left(\prod_{i=1}^{N}\left(\pi_i + O_p\left(\min\left[\tfrac{1}{\sqrt{N}}, \tfrac{1}{\sqrt{T}}\right]\right)\right)\right) == \ln\left(\prod_{i=1}^{N} \pi_i + O_p\left(\min\left[\tfrac{1}{\sqrt{N}}, \tfrac{1}{\sqrt{T}}\right]\right)\right) \tag{8b}$$

that converges to $\ln\left(\prod_{i=1}^{N} \pi_i\right) = \sum_{i=1}^{N} \ln(\pi_i)$ in the limit. Note that $O_p\left(min\left[\tfrac{1}{\sqrt{N}}, \tfrac{1}{\sqrt{T}}\right]\right) = O_p(C_{NT}^{-1})$ and Bai and Ng (2010) indicate that averaging $O_p(C_{NT}^{-1})$ across N produces a term $O_p(C_{NT}^{-2})$ provided that $N/T \to 0$. This implies that $P_m \sim N(0, 1)$ as $T, N \to \infty$ with $N/T \to 0$.

3 Small sample simulations

To examine size and power performance of P and P_m statistics, we consider different data-generating processes (DGPs) with different unit root and co-integration properties. The general form of DGP is given by

$$Z_{it}^{cd} = \mu_i + \Lambda_i{}' F_t + Z_{it} \tag{9}$$

$$F_t = \phi F_{t-1} + u_t \tag{10}$$

$$Z_{it} = \begin{pmatrix} a_{11,i} & a_{12,i} \\ 0 & a_{22,i} \end{pmatrix} Z_{it-1} + \varepsilon_{it} \tag{11}$$

with $\phi = 0.5$, $u_t \sim iidN(0, 1)$, $\varepsilon_{it} \sim iidN(0, \sigma_{\varepsilon i}^2)$, and $\Lambda_i \sim U(0, 0.5)$ for all i where U denotes the uniform distribution. To allow heterogeneous variances across cross sections, $\sigma_{\varepsilon i}^2$ are drawn from $U(0.5, 1.5)$ in line with Dumitrescu and Hurlin (2012) and Emirmahmutoglu and Kose (2011). Since our procedure is invariant to μ_i, we set $\mu_i = 0$ without any loss of generality.[13] For size analysis, $a_{12,i} = 0$ for all i under the null hypothesis. For power analysis, we follow Juodis et al. (2021) and Juodis (2018) and define $a_{12,i} = 0.05 + \xi_i$ with $\xi_i \sim U(-0.1, 0.1)$. Following Arsova and Örsal (2018), the number of factors and the lag order are assumed to be known and set to their true values.[14] The data for $T \in \{50, 100, 200, 300\}$ and $N \in \{10, 15, 25, 50, 100, 200\}$ are generated with T + 50 observations, where the

[13] We are grateful to an anonymous reviewer for this helpful suggestion.

[14] We are grateful to an anonymous reviewer for suggesting (i) to estimate the number of factors and compare the properties of IC criterion of Bai and Ng (2002), ER and GR criteria of Ahn and Horenstein (2013), and the modified IC (IC_m) criterion (outlined in Remark 2) suggested by Huang (2008) for a multivariate framework; and (ii) to investigate whether estimating factors from all $q = 1,..., m$ at the same time, rather than running PCA for each q individually is an important issue in our empirical setting.For (i), we set $r = 3$ in the DGP and estimate the number of factors by using IC, ER, GR, and IC_m criteria with a maximum number of factors $r_{max} = 5$. The results, reported in Table S1 of the On-line Supplement, indicate that all the criteria correctly determine the number of factors for $N = 25$ or greater; and IC_m performs better than

first 50 observations are disregarded to reduce the initial value effect. Monte Carlo simulations are carried out with 5,000 replications at the 5% level.

We concentrate on the performance of P and P_m in the presence of various types of unit root and co-integration in the idiosyncratic components Z_{it}. In that respect, we consider five distinct experiments. We define a full unit root model with $a_{11,i} = a_{22,i} = 1 \forall i$ in experiment 1. We have a co-integrated model with $a_{11,i} = 0.9$ and $a_{22,i} = 1 \forall i$ in experiment 2; and $a_{11,i} = 1$ and $a_{22,i} = 0.9 \forall i$ in experiment 3. Our interest in experiment 4 is centered on a more practical case with different unit root and co-integration properties for different cross sections, defined by $a_{11,i} = a_{22,i} = 1$ for $i \leq N/3$; $a_{11,i} = 0.9$ and $a_{22,i} = 1$ for $N/3 < i \leq 2N/3$; and $a_{11,i} = 1$ and $a_{22,i} = 0.9$ for $2N/3 < i \leq N$. We finally consider the stationary model with $a_{11,i} = a_{22,i} = 0.9 \forall i$ in experiment 5.

We are also interested in comparing the small sample properties of P and P_m statistics with those of the bootstrap test of Dumitrescu and Hurlin (2012) (hereafter DH test) and Emirmahmutoglu and Kose (2011) (hereafter EK test). The bootstrap distributions of DH and EK tests are derived from 1,000 replications, and the bootstrap critical value is obtained at the 5% level. Since DH test requires stationary variables, we employ the first-differenced data for the DGP having a unit root process.

Table 1 presents the size and the size-adjusted power for experiment 1. Note that PC and CA imply P and P_m tests based on the PANIC and PANICCA approaches. The P statistic tends to have size distortions as T is fixed and N grows, and shows better size properties for fixed N with growing T. The P_m statistic appears to have similar size properties to the P statistic, with the advantage of having empirical size close to nominal size as T and N grow together. These results are consistent with the conjecture (outlined above) implying that the P_m statistic is more suitable than the P statistic for the large N panels. In contrast to the P and P_m statistics, the DH and EK tests suffer from size distortions, especially as T grows. The power of P and P_m statistics increases rapidly as N and T grow, which is presumably a reflection of the consistency of the tests. These tests dominate the DH and EK tests, especially in panels where T is large. Finally, while using the PANIC or PANICCA method for de-factorization does not lead to important differences for size properties, the PANIC approach leads to better power properties. The empirical size and the size-adjusted power of the tests for experiments 2–5, reported in Tables 2, 3, 4 and 5, are similar to those for experiment 1.

To sum up, although the P and P_m statistics show mild size distortions for small samples, they have good size and high empirical power as T increases.[15] Dumitrescu and Hurlin (2012) intuitively explain the source of size distortion in panel data. Under the null hypothesis, the individual Wald statistics take large values for some cross-sections. For a given value of N, these large values are not compensated by the

Footnote 14 continued
ER and GR for very low N.For (ii), we re-conduct Monte Carlo simulations by estimating common factors for each q individually based on IC criterion and for all $q = 1,..., m$ based on IC_m criterion. The simulation results, reported in in Tables S2-S6 of On-line Supplement, indicate that size and power properties based on the true factor numbers are similar to those of estimating factors for each q individually and for all $q = 1,..., m$.

[15] Yamada and Toda (1998) also find out that the LA-VAR approach has size distortions for small values of T.

Table 1 Size and power results for experiment 1

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
10	50	0.053	0.071	0.096	0.104	0.082	0.089	0.062	0.107	0.195	0.160	0.173	0.140
	100	0.086	0.071	0.065	0.065	0.058	0.057	0.087	0.239	0.326	0.249	0.295	0.215
	200	0.098	0.090	0.060	0.064	0.055	0.054	0.150	0.458	0.606	0.442	0.569	0.408
	300	0.124	0.128	0.061	0.056	0.053	0.052	0.252	0.590	0.788	0.601	0.761	0.568
15	50	0.064	0.083	0.093	0.099	0.080	0.084	0.053	0.163	0.234	0.197	0.202	0.166
	100	0.073	0.096	0.064	0.069	0.060	0.058	0.078	0.267	0.423	0.331	0.374	0.292
	200	0.111	0.120	0.053	0.060	0.050	0.054	0.170	0.544	0.738	0.599	0.704	0.557
	300	0.151	0.135	0.049	0.050	0.045	0.047	0.238	0.756	0.892	0.768	0.875	0.734
25	50	0.068	0.108	0.119	0.112	0.096	0.095	0.057	0.171	0.309	0.276	0.260	0.231
	100	0.094	0.102	0.082	0.079	0.068	0.068	0.110	0.331	0.552	0.475	0.494	0.416
	200	0.116	0.155	0.064	0.066	0.054	0.057	0.214	0.720	0.884	0.806	0.856	0.766
	300	0.161	0.180	0.063	0.060	0.059	0.054	0.386	0.902	0.975	0.923	0.967	0.905
50	50	0.056	0.127	0.136	0.138	0.106	0.113	0.067	0.267	0.467	0.430	0.394	0.361
	100	0.078	0.134	0.086	0.086	0.074	0.069	0.114	0.534	0.780	0.727	0.726	0.664
	200	0.162	0.196	0.062	0.064	0.056	0.060	0.271	0.902	0.989	0.978	0.984	0.967
	300	0.237	0.311	0.055	0.060	0.050	0.054	0.592	0.989	1.000	0.997	0.999	0.995
100	50	0.088	0.157	0.184	0.192	0.140	0.146	0.077	0.347	0.674	0.642	0.595	0.566

Table 1 (continued)

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
	100	0.110	0.174	0.108	0.108	0.075	0.082	0.149	0.791	0.958	0.939	0.934	0.915
	200	0.233	0.295	0.069	0.069	0.061	0.056	0.451	0.994	1.000	1.000	1.000	1.000
	300	0.352	0.445	0.064	0.062	0.057	0.054	0.775	1.000	1.000	1.000	1.000	1.000
200	50	0.112	0.208	0.275	0.280	0.197	0.208	0.106	0.602	0.896	0.880	0.854	0.836
	100	0.203	0.280	0.130	0.137	0.096	0.099	0.220	0.949	0.998	0.998	0.997	0.996
	200	0.366	0.434	0.079	0.081	0.060	0.060	0.681	1.000	1.000	1.000	1.000	1.000
	300	0.548	0.650	0.071	0.072	0.055	0.059	0.949	1.000	1.000	1.000	1.000	1.000

The DGP is given by $Z_{it}^{cd} = \Lambda_i' F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} a_{11,i} & a_{12,i} \\ 0 & a_{22,i} \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$, $u_t \sim iidN(0,1)$, $\varepsilon_{it} \sim iidN(0, \sigma_{\varepsilon i}^2)$, $\sigma_{\varepsilon i}^2 \sim U(0.5, 1.5)$, and $\Lambda_i \sim U(0, 0.5)$ where U denotes the uniform distribution. $a_{12,i} = 0$ for all i under the null hypothesis for size analysis. $a_{12,i} = 0.05 + \xi_i$ with $\xi_i \sim U(-0.1, 0.1)$ for all i under the alternative hypothesis for power analysis. DH is the bootstrap of Dumitrescu and Hurlin (2012), EK is the bootstrap of Emirmahmutoglu and Kose (2011), and PC (CA) implies that P and P_m tests are based on the PANIC (PANICCA) procedure. Experiment-1: $a_{11,i} = a_{22,i} = 1 \forall i$ (full unit root model)

Table 2 Size and power results for experiment 2

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
10	50	0.062	0.074	0.091	0.091	0.078	0.078	0.088	0.113	0.199	0.179	0.173	0.154
	100	0.056	0.075	0.063	0.065	0.057	0.057	0.194	0.263	0.353	0.302	0.320	0.271
	200	0.068	0.071	0.052	0.051	0.044	0.045	0.346	0.497	0.633	0.541	0.599	0.506
	300	0.088	0.095	0.051	0.055	0.050	0.051	0.549	0.660	0.793	0.705	0.768	0.676
15	50	0.071	0.086	0.087	0.091	0.078	0.078	0.113	0.173	0.257	0.232	0.219	0.198
	100	0.059	0.086	0.068	0.066	0.059	0.059	0.197	0.318	0.424	0.373	0.380	0.329
	200	0.089	0.084	0.053	0.051	0.054	0.050	0.417	0.591	0.743	0.683	0.705	0.642
	300	0.123	0.126	0.053	0.049	0.048	0.044	0.600	0.787	0.895	0.849	0.879	0.824
25	50	0.056	0.096	0.099	0.101	0.082	0.083	0.115	0.216	0.322	0.291	0.273	0.247
	100	0.069	0.080	0.065	0.063	0.057	0.058	0.238	0.378	0.553	0.515	0.499	0.463
	200	0.114	0.115	0.055	0.054	0.051	0.051	0.592	0.761	0.892	0.857	0.868	0.823
	300	0.127	0.140	0.060	0.055	0.058	0.052	0.791	0.913	0.973	0.961	0.966	0.948
50	50	0.055	0.090	0.129	0.133	0.100	0.102	0.114	0.246	0.481	0.448	0.406	0.379
	100	0.069	0.112	0.075	0.074	0.063	0.062	0.336	0.586	0.804	0.772	0.753	0.722
	200	0.117	0.141	0.058	0.057	0.055	0.053	0.810	0.933	0.990	0.987	0.985	0.978
	300	0.131	0.207	0.053	0.054	0.048	0.051	0.970	0.994	1.000	0.999	0.999	0.999
100	50	0.068	0.130	0.165	0.170	0.123	0.128	0.167	0.403	0.706	0.680	0.632	0.606
	100	0.098	0.134	0.085	0.090	0.073	0.072	0.559	0.854	0.956	0.948	0.938	0.928
	200	0.163	0.174	0.066	0.063	0.057	0.055	0.974	0.996	1.000	1.000	1.000	1.000
	300	0.222	0.280	0.058	0.059	0.053	0.055	0.999	1.000	1.000	1.000	1.000	1.000

Table 2 (continued)

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
200	50	0.089	0.151	0.233	0.244	0.170	0.178	0.297	0.620	0.916	0.904	0.875	0.854
	100	0.111	0.185	0.105	0.106	0.076	0.081	0.805	0.971	0.998	0.998	0.998	0.997
	200	0.242	0.264	0.066	0.072	0.055	0.057	0.999	1.000	1.000	1.000	1.000	1.000
	300	0.370	0.437	0.061	0.061	0.051	0.054	1.000	1.000	1.000	1.000	1.000	1.000

The DGP is given by $Z_{it}^{cd} = \Lambda_i' F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} a_{11,i} & a_{12,i} \\ 0 & a_{22,i} \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$, $u_t \sim iidN(0,1)$, $\varepsilon_{it} \sim iidN(0, \sigma_{\varepsilon i}^2)$, $\sigma_{\varepsilon i}^2 \sim U(0.5, 1.5)$, and $\Lambda_i \sim U(0, 0.5)$ where U denotes the uniform distribution. $a_{12,i} = 0$ for all i under the null hypothesis for size analysis. $a_{12,i} = 0.05 + \xi_i$ with $\xi_i \sim U(-0.1, 0.1)$ for all i under the alternative hypothesis for power analysis. DH is the bootstrap of Dumitrescu and Hurlin (2012), EK is the bootstrap of Emirmahmutoglu and Kose (2011), and PC (CA) implies that P and P_m tests are based on the PANIC (PANICCA) procedure. Experiment-2: $a_{11,i} = 0.9$(stationary) and $a_{22,i} = 1$(unit root) $\forall i$

Table 3 Size and power results for experiment 3

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
10	50	0.052	0.058	0.082	0.083	0.069	0.075	0.061	0.107	0.167	0.143	0.146	0.120
	100	0.060	0.109	0.064	0.065	0.058	0.058	0.081	0.207	0.299	0.243	0.270	0.216
	200	0.106	0.129	0.054	0.056	0.047	0.051	0.125	0.380	0.583	0.489	0.551	0.453
	300	0.156	0.179	0.053	0.054	0.048	0.050	0.163	0.607	0.768	0.675	0.740	0.640
15	50	0.055	0.090	0.091	0.091	0.078	0.081	0.070	0.120	0.203	0.179	0.176	0.153
	100	0.079	0.111	0.063	0.065	0.051	0.054	0.084	0.214	0.357	0.308	0.314	0.267
	200	0.095	0.137	0.058	0.057	0.054	0.050	0.152	0.512	0.704	0.638	0.665	0.594
	300	0.157	0.207	0.054	0.054	0.051	0.050	0.225	0.712	0.887	0.825	0.866	0.799
25	50	0.056	0.074	0.101	0.103	0.084	0.085	0.073	0.151	0.269	0.234	0.225	0.194
	100	0.080	0.121	0.064	0.066	0.058	0.059	0.101	0.268	0.487	0.437	0.426	0.382
	200	0.118	0.174	0.061	0.059	0.058	0.054	0.172	0.688	0.853	0.819	0.823	0.781
	300	0.160	0.212	0.058	0.059	0.053	0.053	0.301	0.873	0.966	0.953	0.960	0.940
50	50	0.061	0.108	0.117	0.126	0.090	0.097	0.083	0.202	0.398	0.362	0.329	0.294
	100	0.100	0.140	0.073	0.078	0.059	0.065	0.124	0.456	0.723	0.680	0.662	0.618
	200	0.164	0.234	0.062	0.059	0.055	0.050	0.255	0.880	0.983	0.978	0.975	0.967
	300	0.235	0.391	0.053	0.056	0.051	0.054	0.491	0.982	0.999	1.000	0.999	0.999
100	50	0.081	0.140	0.165	0.173	0.128	0.128	0.084	0.309	0.579	0.550	0.499	0.470
	100	0.109	0.180	0.078	0.083	0.063	0.063	0.191	0.699	0.921	0.908	0.892	0.866
	200	0.257	0.334	0.059	0.062	0.050	0.051	0.452	0.991	1.000	1.000	1.000	0.999
	300	0.383	0.558	0.052	0.054	0.047	0.048	0.619	0.999	1.000	1.000	1.000	1.000

Table 3 (continued)

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
200	50	0.103	0.224	0.218	0.229	0.163	0.166	0.129	0.482	0.817	0.797	0.754	0.730
	100	0.183	0.306	0.107	0.108	0.075	0.077	0.282	0.903	0.995	0.995	0.990	0.989
	200	0.350	0.534	0.070	0.072	0.057	0.059	0.673	1.000	1.000	1.000	1.000	1.000
	300	0.502	0.724	0.058	0.060	0.058	0.056	0.903	1.000	1.000	1.000	1.000	1.000

The DGP is given by $Z_{it}^{cd} = \Lambda_i^{'} F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} a_{11,i} & a_{12,i} \\ 0 & a_{22,i} \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$, $u_t \sim iidN(0,1)$, $\varepsilon_{it} \sim iidN(0, \sigma_{\varepsilon i}^2)$, $\sigma_{\varepsilon i}^2 \sim U(0.5, 1.5)$, and $\Lambda_i \sim U(0, 0.5)$ where U denotes the uniform distribution. $a_{12,i} = 0$ for all i under the null hypothesis for size analysis. $a_{12,i} = 0.05 + \xi_i$ with $\xi_i \sim U(-0.1, 0.1)$ for all i under the alternative hypothesis for power analysis. DH is the bootstrap of Dumitrescu and Hurlin (2012), EK is the bootstrap of Emirmahmutoglu and Kose (2011), and PC (CA) implies that P and P_m tests are based on the PANIC (PANICCA) procedure. Experiment-3: $a_{11,i} = 1$ (unit root) and $a_{22,i} = 0.9$(stationary) $\forall i$.

Table 4 Size and power results for experiment 4

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
10	50	0.043	0.106	0.080	0.087	0.071	0.075	0.078	0.127	0.178	0.154	0.161	0.132
	100	0.060	0.075	0.071	0.067	0.062	0.060	0.128	0.178	0.307	0.244	0.283	0.222
	200	0.076	0.107	0.049	0.053	0.044	0.048	0.176	0.429	0.566	0.450	0.532	0.419
	300	0.137	0.123	0.063	0.053	0.057	0.045	0.299	0.615	0.730	0.615	0.702	0.585
15	50	0.066	0.073	0.096	0.101	0.083	0.086	0.063	0.112	0.225	0.192	0.195	0.164
	100	0.093	0.099	0.065	0.072	0.064	0.066	0.134	0.259	0.393	0.335	0.351	0.297
	200	0.095	0.148	0.061	0.061	0.054	0.055	0.230	0.506	0.697	0.614	0.658	0.575
	300	0.130	0.167	0.055	0.058	0.049	0.051	0.336	0.745	0.839	0.766	0.812	0.736
25	50	0.058	0.085	0.104	0.103	0.081	0.087	0.083	0.160	0.285	0.258	0.242	0.215
	100	0.081	0.099	0.074	0.074	0.063	0.062	0.163	0.321	0.526	0.476	0.476	0.427
	200	0.155	0.157	0.058	0.058	0.053	0.055	0.287	0.649	0.829	0.783	0.796	0.745
	300	0.150	0.194	0.051	0.054	0.050	0.052	0.479	0.859	0.943	0.909	0.930	0.889
50	50	0.067	0.120	0.131	0.129	0.101	0.101	0.114	0.244	0.430	0.401	0.361	0.340
	100	0.102	0.125	0.075	0.076	0.061	0.060	0.166	0.518	0.726	0.696	0.672	0.641
	200	0.135	0.168	0.064	0.065	0.057	0.058	0.473	0.880	0.964	0.955	0.952	0.939
	300	0.271	0.323	0.064	0.063	0.055	0.054	0.677	0.973	0.996	0.993	0.994	0.990
100	50	0.077	0.134	0.167	0.174	0.122	0.133	0.134	0.376	0.637	0.619	0.557	0.541
	100	0.103	0.181	0.087	0.092	0.066	0.067	0.275	0.759	0.906	0.896	0.873	0.858
	200	0.224	0.265	0.068	0.072	0.058	0.062	0.655	0.984	0.999	0.998	0.998	0.996
	300	0.293	0.383	0.051	0.057	0.047	0.052	0.891	0.999	1.000	1.000	1.000	1.000

Table 4 (continued)

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
200	50	0.079	0.234	0.239	0.245	0.178	0.185	0.141	0.539	0.843	0.825	0.785	0.767
	100	0.137	0.240	0.102	0.108	0.075	0.074	0.438	0.906	0.988	0.985	0.980	0.976
	200	0.282	0.410	0.072	0.073	0.059	0.064	0.867	0.999	1.000	1.000	1.000	1.000
	300	0.473	0.587	0.064	0.066	0.051	0.052	0.990	1.000	1.000	1.000	1.000	1.000

The DGP is given by $Z_{it}^{cd} = \Lambda_i' F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} a_{11,i} & a_{12,i} \\ 0 & a_{22,i} \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$, $u_t \sim iidN(0,1)$, $\varepsilon_{it} \sim iidN\left(0, \sigma_{\varepsilon i}^2\right)$, $\sigma_{\varepsilon i}^2 \sim U(0.5, 1.5)$, and $\Lambda_i \sim U(0, 0.5)$ where U denotes the uniform distribution. $a_{12,i} = 0$ for all i under the null hypothesis for size analysis. $a_{12,i} = 0.05 + \xi_i$ with $\xi_i \sim U(-0.1, 0.1)$ for all i under the alternative hypothesis for power analysis. DH is the bootstrap of Dumitrescu and Hurlin (2012), EK is the bootstrap of Emirmahmutoglu and Kose (2011), and PC (CA) implies that P and P_m tests are based on the PANIC (PANICCA) procedure. Experiment-4: $a_{11,i} = a_{22,i} = 1 for i \leq N/3$; $a_{11,i} = 0.9$and $a_{22,i} = 1 for N/3 \leq i < 2N/3$; and $a_{11,i} = 1$ and $a_{22,i} = 0.9$ $for 2N/3 \leq i < N$ (different unit root and co-integration properties for different cross-sections)

Table 5 Size and power results for experiment 5

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
10	50	0.057	0.052	0.085	0.085	0.075	0.072	0.085	0.141	0.205	0.184	0.180	0.161
	100	0.091	0.067	0.063	0.066	0.056	0.056	0.157	0.247	0.335	0.273	0.299	0.247
	200	0.089	0.111	0.058	0.057	0.053	0.049	0.252	0.476	0.619	0.528	0.590	0.493
	300	0.092	0.122	0.053	0.053	0.046	0.048	0.358	0.663	0.793	0.712	0.770	0.690
15	50	0.063	0.096	0.096	0.088	0.080	0.076	0.081	0.154	0.243	0.216	0.206	0.188
	100	0.069	0.104	0.061	0.060	0.055	0.051	0.142	0.306	0.420	0.363	0.372	0.319
	200	0.095	0.107	0.052	0.055	0.044	0.048	0.347	0.578	0.748	0.679	0.710	0.641
	300	0.103	0.127	0.057	0.057	0.051	0.052	0.460	0.745	0.897	0.853	0.879	0.829
25	50	0.051	0.086	0.100	0.102	0.078	0.079	0.095	0.190	0.301	0.280	0.255	0.232
	100	0.074	0.074	0.066	0.066	0.058	0.059	0.197	0.374	0.563	0.512	0.504	0.462
	200	0.105	0.141	0.055	0.056	0.049	0.050	0.422	0.758	0.893	0.864	0.864	0.828
	300	0.133	0.195	0.058	0.054	0.049	0.047	0.626	0.911	0.977	0.966	0.972	0.955
50	50	0.074	0.096	0.119	0.128	0.097	0.107	0.115	0.254	0.463	0.440	0.395	0.372
	100	0.089	0.088	0.080	0.078	0.066	0.062	0.266	0.540	0.794	0.761	0.745	0.704
	200	0.145	0.170	0.061	0.059	0.056	0.057	0.603	0.917	0.988	0.985	0.984	0.979
	300	0.179	0.222	0.057	0.059	0.052	0.053	0.834	0.994	1.000	1.000	1.000	0.999
100	50	0.071	0.119	0.166	0.169	0.126	0.128	0.178	0.394	0.687	0.664	0.610	0.583
	100	0.097	0.150	0.084	0.085	0.067	0.067	0.478	0.807	0.962	0.952	0.940	0.927
	200	0.156	0.209	0.062	0.060	0.051	0.054	0.822	0.995	1.000	1.000	1.000	1.000
	300	0.284	0.336	0.051	0.053	0.048	0.048	0.970	1.000	1.000	1.000	1.000	1.000

Table 5 (continued)

N	T	Size						Power					
		DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$	DH	EK	P_{PC}	P_{CA}	$P_{m,PC}$	$P_{m,CA}$
200	50	0.089	0.177	0.233	0.238	0.168	0.171	0.274	0.589	0.907	0.887	0.860	0.839
	100	0.179	0.234	0.098	0.105	0.072	0.077	0.713	0.959	0.999	0.999	0.999	0.998
	200	0.252	0.377	0.066	0.068	0.053	0.056	0.975	1.000	1.000	1.000	1.000	1.000
	300	0.431	0.534	0.063	0.066	0.056	0.055	1.000	1.000	1.000	1.000	1.000	1.000

The DGP is given by $Z_{it}^{cd} = \Lambda_i' F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} a_{11,i} & a_{12,i} \\ 0 & a_{22,i} \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$, $u_t \sim iidN(0,1)$, $\varepsilon_{it} \sim iidN\left(0, \sigma_{\varepsilon i}^2\right)$, $\sigma_{\varepsilon i}^2 \sim U(0.5, 1.5)$, and $\Lambda_i \sim U(0, 0.5)$ where U denotes the uniform distribution. $a_{12,i} = 0$ for all i under the null hypothesis for size analysis. $a_{12,i} = 0.05 + \xi_i$ with $\xi_i \sim U(-0.1, 0.1)$ for all i under the alternative hypothesis for power analysis. DH is the bootstrap of Dumitrescu and Hurlin (2012), EK is the bootstrap of Emirmahmutoglu and Kose (2011), and PC (CA) implies that P and P_m tests are based on the PANIC (PANICCA) procedure. Experiment-5: $a_{11,i} = a_{22,i} = 0.9 \forall i$(stationary model)

realizations obtained for other cross sections since the latter only range from zero to the chi-squared critical value. As a result, the panel statistic tends to be larger than the corresponding critical value. As N increases, the probability of obtaining large values for some cross sections increases, and the panel test tends to over-reject the null hypothesis for small T values. In that respect, the solution of Juodis et al. (2021) based on the Split Panel Jackknife procedure leads the Wald statistic to have good size and power properties even with a moderate T dimension in large N panels.

Another insightful consequence of our simulations is that the DH and EK bootstrap-based tests tend to reject the null hypothesis as T increases.[16] Using the PANIC and PANICCA methods seems more convenient than using the bootstrap method in modeling cross-sectional dependence for testing Granger causality in heterogeneous panels with a factor structure.[17]

4 Empirical application

There is a huge and growing empirical literature on causality[18] between export and economic growth. The literature mainly focuses on the export-led growth (ELG) and growth-led export (GLE) hypotheses. The ELG hypothesis implies that export activity leads to economic growth. The GLE hypothesis postulates a reverse direction and implies causality from economic growth to export. On the other hand, if there is bidirectional causality between export and economic growth, this supports the existence of feedback relation.

The ELG and GLE hypotheses are examined by a large number of empirical studies. Giles and Williams (2000) review more than 150 empirical papers published until the late 1990s. Most of these papers use time series techniques and do not find any consistent result for the validity of the ELG and GLE hypotheses. After the 2000s, the empirical literature focuses on using panel causality methods to benefit from the high-power property of such tests in small samples. Initial papers assume homogeneity and independency across countries (Hsiao and Hsiao 2006; Won and Hsiao 2008; Çetintaş and Barişik 2009; Narayan and Smyth 2009; Nasreen and Anwar 2014). Some recent studies allow heterogeneity but do not consider cross-sectional dependence across countries (Sanjuán-López and Dawson 2010; Pradhan and Bagchi 2012; Dawson and Sanjuán-López 2013). Some other recent papers, among others Tekin (2012) and Chang et al. (2013), find out that allowing heterogeneity and cross-sectional dependence (with bootstrapping) is important to examine the direction of causality between export and economic growth. Nonetheless, to the best of our knowledge, no study accounts for cross-sectional dependence based on factor models.

[16] We are grateful to an anonymous reviewer for suggesting to clarify that our setup for the dependency structure, based on factor model, differs from that of Dumitrescu and Hurlin (2012) wherein the dependence takes the form of a correlation coefficient in residuals ε_{it} equal to 0.5.

[17] We are grateful to an anonymous reviewer for raising out that Dumitrescu and Hurlin (2012) and Emirmahmutoglu and Kose (2011) do not provide theoretical justifications for their bootstrap procedures to perform better in a panel VAR framework with a factor structure.

[18] We are grateful to an anonymous reviewer for suggesting to clarify that causality hereafter refers to Granger causality for the sake of consistency.

This gap in the literature motivates us to question whether using factor models makes sense for causality between export and growth for OECD countries, in which international trade plays a crucial role in economic structure.[19] For a comparison purpose, we also conduct the existing approaches proposed by Kónya (2006), Dumitrescu and Hurlin (2012), and Emirmahmutoglu and Kose (2011).[20] Our data covers the real GDP and export for 20 OECD countries from 1996:Q1 to 2019:Q4.[21] The real GDP is constructed using the GDP deflator, and the real export is obtained using the consumer price index. The data are retrieved from the International Financial Statistics Database. The natural logarithm of the variables is employed in the estimations.

With respect to the panel causality literature focusing on OECD countries, Kónya (2006) examines causality between real export and real gross domestic product (GDP) in 24 OECD countries for the period from 1960 to 1997. By conducting the panel causality test on the SUR, he finds one-way causality from export to GDP in Belgium, Denmark, Iceland, Ireland, Italy, New Zealand, Spain, and Sweden; one-way causality from GDP to export in Austria, France, Greece, Japan, Mexico, Norway, and Portugal; two-way causality between exports and growth in Canada, Finland, and the Netherlands; no causality in the case of Australia, Luxembourg, S. Korea, Switzerland, the UK, and the USA. Emirmahmutoglu and Kose (2011) use the panel causality test based on the LA-VAR approach to investigate causality between export and growth in 20 OECD countries from 1987 to 2006. Different from Kónya (2006), they show one-way causality from export to growth in Japan, from growth to export in Australia, Germany, Norway, S. Korea, and the USA, and two-way causality in Türkiye. They moreover support the validity of the GLE hypothesis for OECD countries. Our study differs from these studies by accounting for cross-sectional dependence and demonstrates the importance of factor modeling framework for country-specific results.

In the LA-VAR approach, we need to determine the maximum integration degree (d) of the variables for each cross section. To test for unit root in panel data, one important issue is to consider cross-sectional dependence. As a preliminary analysis, we first test for cross-sectional dependence by using LM (Lagrange Multiplier) test proposed by Breusch and Pagan (1980), and CD_{LM} and CD tests developed by Pesaran (2021). The results in Table 6 indicate that the null hypothesis of cross-sectional independence is strongly rejected for both export and GDP, supporting the evidence on cross-sectional dependence. We benefit from the PANIC approach of Bai and Ng (2004) and the PANICCA approach of Reese and Westerlund (2016) to investigate the

[19] With increasing globalization, policy makers debate many issues on trade in OECD countries, such as trade liberalization, technology transfer policies, productivity, etc. in the last decades. By reducing many barriers to import and export, OECD members aim to boost global trade, spur economic growth, and create more and jobs (Benz and Jaax 2020). Since OECD countries are highly integrated with each other, a shock in one country is expected to be easily transmitted to other countries through intensive economic interrelationships. Nonetheless, these countries have their own characteristics stemming from differences in economic structure and trade specialization as well as in factor endowments which lead to different levels of economic development.

[20] User-friendly codes are available for these existing approaches in GAUSS Time Series and Panel Data library (TSPDlib) of Nazlioglu (2021), publicly available at: https://github.com/aptech/tspdlib.

[21] It is worthwhile noting that the Monte Carlo results (presented in Sect. 3) for the sample with N = 25 and T = 100, corresponding to our sample with N = 20 and T = 96, indicate that the P and P_m tests based on the PANIC procedure works better than other tests considered for small sample comparisons.

Table 6 Results from the cross-sectional dependence tests

Test	GDP		Export	
	Statistic	*p*-value	Statistic	*p*-value
LM	1438.314***	0.000	2180.511***	0.000
CD_{LM}	64.037***	0.000	102.111***	0.000
CD	31.542***	0.000	40.670***	0.000

LM is the cross-sectional dependence test with chi-square distribution with N(N− 1)/2 degrees of freedom developed by Breusch and Pagan (1980). CD_{LM} and CD tests are the cross-sectional dependence tests with standard normal distribution developed by Pesaran (2021). The tests are based on the residuals from the ADF regressions with constant term. The optimal lag(s) in the ADF regressions are selected by Schwarz information criterion by setting maximum lags to 4. ***Denotes the statistical significance at 1%

unit root properties of the series. The results from the unit root analysis,[22] reported in Table 7 indicate that the maximum order of integration is determined as one for all countries.

We estimate common factors by means of the PANIC and the PANICCA approaches for de-factoring the data. To determine the number of common factors in the PANIC approach, we benefit from three different information criteria, IC panel information criterion of Bai and Ng (2002), and ER (eigenvalue ratio) and GR (growth ratio) criteria of Ahn and Horenstein (2013). These criteria are based on PCA that often requires standardizing panel data before estimating unobserved common factors. Greenaway-McGrevy et al. (2012) investigate the importance of standardization in eigenvalue-based decompositions by particularly focusing on PCA. They find that the validity of standardization is dependent on the main source of heteroscedasticity. If heteroscedasticity arises from idiosyncratic errors, standardization is required to consistently estimate PCA factors and factor numbers. If heteroscedasticity comes from factor loadings, standardization may lead to inconsistent estimation of PCA factors and over-estimating factor numbers. Since it is difficult to know the source of heteroscedasticity in practice, Greenaway-McGrevy et al. (2012) suggest using a minimum rule, whereby the factor number is estimated as $\widehat{r}_{min} = \min[\widehat{r}_{std}, \widehat{r}_{no-std}]$ where $\widehat{r}_{std}$ and $\widehat{r}_{no-std}$, respectively, denote the estimated number of factors with and without standardization.

Table 8 reports the results from the factor number estimations from the panel comprising export and GDP for all units. The number of factors is estimated by setting the maximum number $r_{max} = 3$. Since the variables have the unit process, we conduct PCA on the first-differenced data to avoid spurious results arising from non-stationarity. While the IC, ER, and GR criteria select different factor numbers (3, 1, and 2) for the non-standardized data, they chose one common factor for the standardized data. Accordingly, the minimum rule of Greenaway-McGrevy et al. (2012) determines one factor based on all three criteria.[23] Figure 1 plots the estimated factor from both PCA and CA. At the first glance, the factor estimates from both approaches almost coincide,

[22] The unit root tests were conducted with the TSPDlib.

[23] The first factor from PCA explains almost 50% of total variation.

Table 7 Results from unit root tests

	PANIC				PANICCA			
	Level		First difference		Level		First difference	
	ADF	*p*-value	ADF	*p*-value	ADF	*p*-value	ADF	*p*-value
GDP								
Australia	2.315	0.995	− 7.428***	0.000	4.240	1.000	− 4.243***	0.000
Austria	0.424	0.797	− 12.013***	0.000	− 0.453	0.510	− 4.141***	0.000
Canada	− 0.920	0.315	− 8.779***	0.000	− 0.157	0.620	− 9.035***	0.000
Denmark	− 0.711	0.403	− 12.448***	0.000	− 0.674	0.420	− 12.510***	0.000
Finland	− 1.281	0.182	− 10.484***	0.000	0.825	0.885	− 11.331***	0.000
France	− 0.214	0.600	− 10.899***	0.000	0.138	0.718	− 6.547***	0.000
Germany	− 0.329	0.557	− 9.451***	0.000	0.411	0.795	− 10.115***	0.000
Italy	2.000	0.988	− 2.742***	0.007	1.072	0.922	− 4.394***	0.000
Japan	0.004	0.675	− 9.587***	0.000	0.450	0.805	− 9.687***	0.000
Mexico	− 0.807	0.362	− 7.533***	0.000	− 1.429	0.142	− 7.165***	0.000
Netherlands	0.149	0.720	− 8.857***	0.000	− 0.847	0.345	− 9.000***	0.000
New Zealand	− 1.381	0.155	− 12.025***	0.000	2.203	0.993	− 8.387***	0.000
Norway	− 4.668***	0.000	− 9.652***	0.000	− 0.721	0.398	− 9.393***	0.000
Portugal	− 1.320	0.172	− 4.201***	0.000	− 1.030	0.270	− 3.901***	0.000
S. Korea	1.839	0.983	− 7.559***	0.000	1.719	0.978	− 7.543***	0.000
Spain	− 1.207	0.207	− 2.763***	0.007	− 1.330	0.170	− 1.961**	0.048
Sweden	− 1.421	0.145	− 10.390***	0.000	0.313	0.767	− 8.649***	0.000
Türkiye	1.379	0.955	− 8.071***	0.000	1.323	0.950	− 3.823***	0.000

Table 7 (continued)

	PANIC				PANICCA			
	Level		First difference		Level		First difference	
	ADF	*p*-value	ADF	*p*-value	ADF	*p*-value	ADF	*p*-value
UK	− 1.148	0.228	− 9.362***	0.000	0.471	0.810	− 10.124***	0.000
USA	0.668	0.855	− 10.730***	0.000	2.420	0.995	− 4.535***	0.000
Export								
Australia	− 1.471	0.133	− 8.489***	0.000	− 2.639***	0.009	− 8.069***	0.000
Austria	1.227	0.940	− 8.283***	0.000	0.485	0.813	− 8.288***	0.000
Canada	0.201	0.735	− 9.230***	0.000	0.608	0.840	− 3.410***	0.000
Denmark	− 0.270	0.578	− 13.838***	0.000	− 2.297**	0.022	− 13.594***	0.000
Finland	− 0.008	0.670	− 17.215***	0.000	0.663	0.853	− 16.423***	0.000
France	0.211	0.740	− 9.359***	0.000	− 0.752	0.385	− 9.751***	0.000
Germany	1.135	0.930	− 9.683***	0.000	0.293	0.762	− 10.716***	0.000
Italy	0.557	0.830	− 9.235***	0.000	1.136	0.930	− 8.834***	0.000
Japan	0.726	0.865	− 8.660***	0.000	0.956	0.905	− 8.951***	0.000
Mexico	0.127	0.713	− 8.694***	0.000	− 0.197	0.605	− 8.773***	0.000
Netherlands	0.391	0.790	− 12.977***	0.000	− 1.538	0.117	− 12.805***	0.000
New Zealand	− 0.709	0.403	− 10.656***	0.000	− 1.196	0.210	− 10.492***	0.000
Norway	0.470	0.810	− 7.668***	0.000	1.015	0.915	− 8.038***	0.000
Portugal	1.798	0.983	− 12.826***	0.000	− 0.317	0.563	− 12.597***	0.000

Table 7 (continued)

	PANIC				PANICCA			
	Level		First difference		Level		First difference	
	ADF	*p*-value	ADF	*p*-value	ADF	*p*-value	ADF	*p*-value
S. Korea	− 3.053***	0.003	− 8.251***	0.000	− 0.915	0.318	− 9.477***	0.000
Spain	0.970	0.907	− 4.956***	0.000	0.572	0.833	− 10.658***	0.000
Sweden	1.269	0.945	− 10.185***	0.000	0.678	0.858	− 10.598***	0.000
Türkiye	− 0.190	0.608	− 9.395***	0.000	− 1.485	0.128	− 8.444***	0.000
UK	0.249	0.750	− 10.530***	0.000	− 0.859	0.340	− 10.202***	0.000
USA	− 0.824	0.355	− 8.168***	0.000	− 0.581	0.458	− 7.848***	0.000

ADF is the augmented Dickey and Fuller (1979) unit root test. The tests are based on the model with constant term. The optimal lag(s) are selected by Schwarz information criterion by setting maximum lags to 4 in the ADF regressions. The *p*-values are based on the look-up table of Bai and Ng (2004). The PANIC is based on the principal components analysis (PCA). The optimal number of factors are selected by the IC_2 panel information criterion of Bai and Ng (2002) by setting maximum number of factors to 3. ***,**,*Denote the statistical significance at 1, 5, and 10%, respectively.

Table 8 Estimated number of factors

Non-standardized data			Standardized data		
IC	ER	GR	IC	ER	GR
3	1	2	1	1	1

Maximum number of factors $r_{max} = 3$. IC denotes IC_2 criterion of Bai and Ng (2002), ER and GR respectively denote the eigen value ratio and the growth ratio of Ahn and Horenstein (2013)

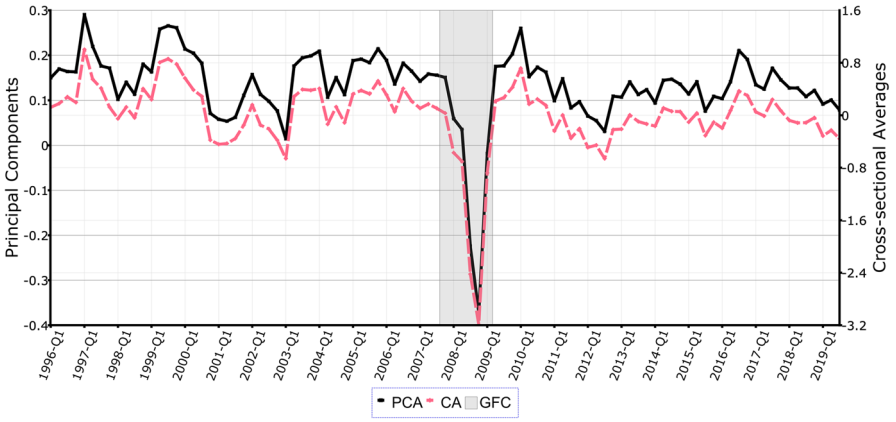


Fig. 1 Estimated common factors. PCA (principal components, solid line, left axes), CA (cross-sectional averages, dash line, and right axes), GFC (the 2007/2009 Global Financial Crisis)

and there is a downfall midway through the sample, which can be attributed to the Global Financial Crisis between mid 2007 and early 2009. Except for the downfall, both PCA and CA estimates look like to be circling on a constant mean. Hence, the estimated factor from either approach captures fluctuations in the data.

The results from the panel LA-VAR causality test with the PANIC and PANICCA approaches are presented in Tables 9 and 10, respectively.[24] The panel statistics, P and P_m, indicate that the null hypothesis of Granger non-causality is rejected, supporting the existence of both the ELG and GLE hypotheses in OECD countries. The rejection of the joint null hypothesis for the panel does not mean that all countries have causality since the alternative hypothesis allows some cross sections not to have non-causality.

We proceed with the cross-section specific results to examine the countries having causality and non-causality. We should, at this point, note that imposing the null hypothesis for 20 countries means 40 tests for causality, which naturally raises a concern about mass significance. Using a p-value correction method may be of critical

[24] After removing the cross-sectional dependence from the data, we estimate the LA-VAR $(k_i + d_i)$ models with the optimal lag (k_i) based on Schwarz information criterion and the maximum integration degree of the variables (d_i) as one.

Table 9 Results from the panel LA-VAR causality test with PANIC

	H_0: Export $\neq>$ GDP		H_0: GDP $\neq>$ Export		Lags
	Wald	p-value	Wald	p-value	
Country-specific results					
Australia	2.918	1.000	20.321**	0.000	2
Austria	0.031	1.000	0.757	1.000	1
Canada	4.088	0.774	5.525	0.323	1
Denmark	0.848	1.000	2.205	1.000	1
Finland	4.665	1.000	0.159	1.000	2
France	0.744	1.000	17.298**	0.000	1
Germany	0.463	1.000	0.533	1.000	1
Italy	0.985	1.000	1.687	1.000	1
Japan	0.639	1.000	0.494	1.000	1
Mexico	0.173	1.000	0.444	1.000	1
Netherlands	0.007	1.000	0.863	1.000	1
New Zealand	0.017	1.000	0.495	1.000	1
Norway	0.770	1.000	12.532**	0.036	2
Portugal	3.151	1.000	1.581	1.000	1
S. Korea	8.549	0.266	6.072	0.768	2
Spain	2.795	1.000	6.751	1.000	3
Sweden	0.007	1.000	1.882	1.000	1
Türkiye	1.055	1.000	2.106	1.000	2
UK	1.372	1.000	1.578	1.000	1
USA	15.120**	0.020	0.897	1.000	2
Panel results					
P	59.988**	0.022	102.657**	0.000	
P_m	2.235**	0.013	7.005**	0.000	
# of factor	1		1		

$\neq>$ Denotes non-Granger causality hypothesis. The panel LA-VAR model is defined as $y_{it} = \sum_{j=1}^{k_i+d_i} \alpha_{ij} y_{i,t-j} + \sum_{j=1}^{k_i+d_i} \beta_{ij} x_{i,t-j} + \varepsilon_{it}$. The LA-VAR models are estimated by using the level of the variables with $d_i = 1 \forall i$. The optimal lag(s), k_i, are selected by Schwarz information criterion by setting maximum lags to 4 in VAR model. The PANIC is based on the principal components analysis (PCA). The optimal number of factors are selected by IC_2 panel information criterion of Bai and Ng (2002) by setting maximum numbers to 3. The factors are estimated from the first-differenced and standardized data. For panel results, the p-values are based on the asymptotic distribution. For country-specific results, the p-values are the adjusted p-values based on the Holm's procedure. **Denotes the statistical significance at 5%

Table 10 Results from the panel LA-VAR causality test with PANICCA

	H_0: Export $\neq$> GDP		H_0: GDP $\neq$> Export		Lags
	Wald	*p*-value	Wald	*p*-value	
Country-specific results					
Australia	2.950	1.000	21.268**	0.000	2
Austria	0.033	1.000	0.386	1.000	1
Canada	4.345	0.666	4.868	0.459	1
Denmark	0.548	1.000	2.299	1.000	1
Finland	3.208	1.000	0.129	1.000	2
France	0.881	1.000	12.338**	0.000	1
Germany	0.273	1.000	0.322	1.000	1
Italy	0.584	1.000	0.903	1.000	1
Japan	0.439	1.000	0.459	1.000	1
Mexico	0.221	1.000	0.476	1.000	1
Netherlands	0.008	1.000	1.253	1.000	1
New Zealand	0.000	1.000	0.333	1.000	1
Norway	0.453	1.000	12.237**	0.036	2
Portugal	2.972	1.000	1.330	1.000	1
S. Korea	5.164	0.437	0.527	1.000	1
Spain	3.136	1.000	2.176	1.000	3
Sweden	0.177	1.000	2.586	1.000	1
Türkiye	0.951	1.000	2.366	1.000	2
UK	1.764	1.000	1.848	1.000	1
USA	18.515**	0.000	1.085	1.000	2
Panel results					
P	61.484**	0.016	89.397**	0.000	
P_m	2.402**	0.008	5.523**	0.000	

$\neq$> Denotes non-Granger causality hypothesis. The panel LA-VAR model is defined as $y_{it} = \sum_{j=1}^{k_i+d_i} \alpha_{ij} y_{i,t-j} + \sum_{j=1}^{k_i+d_i} \beta_{ij} x_{i,t-j} + \varepsilon_{it}$. The LA-VAR models are estimated by using the level of the variables with $d_i = 1 \forall i$. The optimal lag(s), k_i, are selected by Schwarz information criterion by setting maximum lags to 4 in VAR model. The PANICCA is based on the cross-sectional averages of the variables. For panel results, the *p*-values are based on the asymptotic distribution. For country-specific results, the *p*-values are the adjusted *p*-values based on the Holm's procedure. **Denotes the statistical significance at 5%

importance in the case of relatively large numbers of tests,[25] and we adjust p-values using the Holm's (1979) procedure. Our discussion hereafter is based on the adjusted p-value at 5%, that an adjusted p-value less than 0.05 rejects the null hypothesis of non-causality. The country-specific results from the PANIC and PANICCA approaches support the ELG hypothesis for USA; the GLE hypothesis for Australia, France, and Norway; and the neutrality for the majority of countries (Austria, Canada, Denmark, Finland, Germany, Italy, Japan, Mexico, Netherlands, New Zealand, Portugal, S. Korea, Spain, Sweden, Türkiye, and UK).

The results from the panel SUR causality test of Kónya (2006)[26] are given in Table 11. In this approach, testing for the cross-sectional dependence is crucial for selecting the appropriate estimator since the SUR estimator is more efficient than the OLS only if there is cross-sectional dependence in the system. To investigate the existence of cross-sectional dependence, we carry out the cross-sectional dependence tests on the residuals from the individual OLS regressions. The cross-sectional dependence tests reject the null of cross-sectional independence, implying that the SUR estimator is appropriate rather than country-by-country OLS estimations. The results support the ELG hypothesis for Netherlands; the GLE hypothesis for Mexico and UK; and the neutrality for Australia, Austria, Canada, Denmark, Finland, France, Germany, Italy, Japan, New Zealand, Norway, Portugal, S. Korea, Spain, Sweden, Türkiye and USA.

The results from the panel VAR causality test of Dumitrescu and Hurlin (2012)[27] are reported in Table 12. Testing for the cross-sectional dependence is also required to decide whether one needs to use asymptotic or bootstrap distribution since the asymptotic distribution is only valid under cross-sectional independence. The cross-sectional dependence tests indicate that the null of cross-sectional independence is strongly rejected, implying that we need to use bootstrap distribution. The results show that the panel statistic is larger than the bootstrap critical values; hence, the null hypothesis of non-causality for the panel is rejected. This finding supports two-way causality between export and GDP, implying the existence of feedback relation. The country-specific results provide evidence on the validity of the ELG hypothesis for Austria; the GLE hypothesis for Denmark, Finland, Italy, and Sweden; the feedback relation in S. Korea; and the neutrality for Australia, Canada, France, Germany, Japan, Mexico, Netherlands, New Zealand, Norway, Portugal, Spain, Türkiye, UK, and USA.

The results from the panel LA-VAR causality test of Emirmahmutoglu and Kose (2011)[28] are reported in Table 13. We test the cross-sectional dependence since the

[25] We are grateful to an anonymous reviewer for raising this point, with a helpful suggestion to employ the Holm's p-value correction approach for the issue of mass significance. This problem is already recognized in the literature on testing for panel unit root and co-integration (see, among others, Hanck 2013; Arsova and Örsal 2020).

[26] Following Kónya (2006), we determine the optimal number of lags by allowing maximal lags to differ across variables but to be the same across equations. We accordingly estimate the SUR system by setting maximum lags to four, and then choose the optimal lag with Schwarz information criterion.

[27] Note that since this method requires using stationary variables, we estimate the VAR (k_i) models, with the optimal lag (k_i) based on Schwarz information criterion from maximum four lags, in first differences because export and GDP have unit root in levels and are stationary in first-differences.

[28] We estimate the LA-VAR ($k_i + d_i$) models in levels with the optimal lag (k_i) based on Schwarz information criterion and the maximum integration degree of the variables (d_i) as one for each country based on the results from the unit root analysis in Table 7.

Table 11 Results from the panel SUR causality test

	H_0: Export $\neq>$ GDP		H_0: GDP $\neq>$ Export	
	Wald	*p*-value	Wald	*p*-value
Country-specific results				
Australia	0.002	1.000	6.604	0.108
Austria	4.810	0.136	0.262	1.000
Canada	3.630	1.000	0.990	1.000
Denmark	0.267	1.000	4.728	0.180
Finland	1.267	1.000	3.056	0.484
France	4.387	0.448	1.971	1.000
Germany	0.305	1.000	0.043	1.000
Italy	3.496	1.000	0.000	1.000
Japan	0.768	1.000	4.329	0.429
Mexico	6.042	0.270	9.428**	0.019
Netherlands	15.021**	0.000	3.148	0.429
New Zealand	0.086	1.000	5.840	0.136
Norway	0.356	1.000	0.091	1.000
Portugal	4.569	0.144	0.432	1.000
S. Korea	2.233	1.000	4.141	0.136
Spain	9.829	0.114	1.839	1.000
Sweden	0.380	1.000	5.860	0.182
Türkiye	0.187	1.000	2.134	0.790
UK	7.044	0.126	15.014**	0.000
USA	0.041	1.000	0.109	1.000

Cross-sectional dependence tests

	Statistic	*p*-value
LM	4877.677**	0.000
CD_{LM}	240.473**	0.000
CD	47.874**	0.000

$\neq>$ Denotes non-Granger causality hypothesis. LM is the cross-sectional dependence test with chi-square distribution with $N(N-1)/2$ degrees of freedom developed by Breusch and Pagan (1980). CD_{LM} and CD tests are the cross-sectional dependence tests with standard normal distribution developed by Pesaran (2021). For cross-sectional dependence tests, the *p*-values are based on the asymptotic distribution. The *p*-values are the adjusted *p*-values based on the Holm's procedure. **Denotes the statistical significance at 5%

limiting distribution of the panel statistic is no longer valid in the existence of cross-sectional dependence. The results from the cross-sectional dependence tests show that the null of cross-sectional independence is strongly rejected, implying that we need to use the bootstrap critical values. The panel statistic is greater than the bootstrap critical values and rejects the null hypothesis of non-causality for the panel, supporting the existence of both the ELG and GLE hypotheses. The country-specific results imply the existence of the ELG hypothesis for S. Korea; the GLE hypothesis for Mexico and Sweden; the feedback relation for Finland; and the neutrality for Australia, Austria, Canada, Denmark, France, Germany, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Türkiye, UK, and USA.

To succinctly compare the findings from all the panel causality tests carried out so far, we summarize the results in Table 14. The panel results support the validity

Table 12 Results from the panel VAR causality test

	H_0: Export $\neq>$ GDP		$H_0 : GDP \neq> Export$		Lags
	Wald	p-value	Wald	p-value	
Country-specific results					
Australia	0.646	1.000	2.950	0.946	1
Austria	11.376**	0.019	6.777	0.126	1
Canada	8.595	0.054	0.166	1.000	1
Denmark	6.992	0.420	19.384**	0.000	2
Finland	2.146	1.000	14.636**	0.000	1
France	0.884	1.000	0.320	1.000	1
Germany	7.332	0.119	1.567	1.000	1
Italy	0.917	1.000	10.579**	0.018	1
Japan	2.000	1.000	0.390	1.000	1
Mexico	0.461	1.000	4.283	0.468	1
Netherlands	3.698	0.702	6.227	0.169	1
New Zealand	2.062	1.000	0.080	1.000	1
Norway	5.619	0.288	2.907	0.946	1
Portugal	1.615	1.000	2.641	1.000	2
S. Korea	14.195**	0.000	8.583**	0.048	1
Spain	7.522	0.345	10.449	0.075	2
Sweden	2.613	1.000	10.685**	0.018	1
Türkiye	3.668	0.702	0.009	1.000	1
UK	0.864	1.000	1.516	1.000	1
USA	4.134	1.000	2.793	1.000	2
Panel results					
Panel statistic	9.142**		11.971**		
Bootstrap critical value (5%)	2.021		1.662		

Table 12 (continued)

Cross-sectional dependence tests		
	Statistic	*p*-val
LM	4009.129**	0.000
CD_{LM}	195.917**	0.000
CD	27.945**	0.000

$\neq>$ Denotes non-Granger causality null hypothesis. The panel VAR model is defined as $y_{it} = \mu_i + \sum_{j=1}^{k_i} \alpha_{ij} y_{i,t-j} + \sum_{j=1}^{k_i} \beta_{ij} x_{i,t-j} + v_{it}$. The VAR models are estimated by using the first difference of the variables. The optimal lag(s), k_i, are selected by Schwarz information criterion by setting maximum lags to 4 in VAR models. The bootstrap critical values are obtained from 1000 replications. See Eq. (33) in Dumitrescu and Hurlin (2012, p. 1459) for the definition of the panel statistic. LM is the cross-sectional dependence test with chi-square distribution with $N(N-1)/2$ degrees of freedom developed by Breusch and Pagan (1980). CD_{LM} and CD tests are the cross-sectional dependence tests with standard normal distribution developed by Pesaran (2004, 2021). The cross-sectional dependence tests are calculated by using the estimated residuals $\hat{v}_{it}$ For cross-sectional dependence tests, the *p*-values are based on the asymptotic distribution. For country-specific results, the *p*-values are the adjusted *p*-values based on the Holm's procedure. **Denotes the statistical significance at 5%

Table 13 Results from the panel LA-VAR causality test

	H_0: Export $\neq>$ GDP		H_0: GDP $\neq>$ Export		Lags
	Wald	*p*-value	Wald	*p*-value	
Country-specific results					
Australia	0.710	1.000	5.258	0.864	2
Austria	4.319	1.000	10.113	0.096	2
Canada	7.987	0.090	0.072	1.000	1
Denmark	2.296	1.000	0.711	1.000	1
Finland	18.107**	0.000	17.837**	0.000	3
France	3.586	1.000	3.717	1.000	2
Germany	5.418	1.000	3.844	1.000	2
Italy	2.515	1.000	7.289	0.338	2
Japan	0.905	1.000	0.734	1.000	1
Mexico	3.243	1.000	9.227**	0.038	1
Netherlands	7.499	0.408	7.511	0.322	2
New Zealand	1.866	1.000	0.087	1.000	1
Norway	5.761	0.896	3.268	1.000	2
Portugal	0.043	1.000	0.981	1.000	1
S. Korea	15.626**	0.000	10.409	0.085	2

Table 13 (continued)

	H_0: Export $\neq>$ GDP		H_0: GDP $\neq>$ Export		Lags
	Wald	p-value	Wald	p-value	
Spain	6.838	1.000	11.643	0.135	3
Sweden	0.427	1.000	12.383**	0.038	2
Türkiye	1.333	1.000	0.728	1.000	2
UK	0.986	1.000	1.254	1.000	1
USA	3.473	1.000	2.755	1.000	2
Panel results					
Panel statistic	99.720**		113.797**		
Bootstrap critical value (5%)	55.623		55.972		

Cross-sectional dependence tests

	Statistic	p-value
LM	7601.739**	0.000
CD_{LM}	380.214**	0.000
CD	132.205**	0.000

$\neq>$ Denotes non-Granger causality hypothesis. The panel LA-VAR model is defined as $y_{it} = \mu_i + \sum_{j=1}^{k_i+d_i} \alpha_{ij} y_{i,t-j} + \sum_{j=1}^{k_i+d_i} \beta_{ij} x_{i,t-j} + \varepsilon_{it}$. The LA-VAR models are estimated by using the level of the variables with $d_i = 1 \forall i$. The optimal lag(s), k_i, are selected by Schwarz information criterion by setting maximum lags to 4 in VAR model. The bootstrap critical values are obtained from 1,000 replications. See Eq. (9) in Emirmahmutoglu and Kose (2011, p. 872) for the definition of the panel statistic. LM is the cross-sectional dependence test with chi-square distribution with $\text{N(N} - 1)/2$ degrees of freedom developed by Breusch and Pagan (1980). CD_{LM} and CD tests are the cross-sectional dependence tests with standard normal distribution developed by Pesaran (2004, 2021). The cross-sectional dependence tests are calculated by using the estimated residuals $\hat{\varepsilon}_{it}$. For cross-sectional dependence tests, the p-values are based on the asymptotic distribution. For country-specific results, the p-values are the adjusted p-values based on the Holm's procedure. **Denotes the statistical significance at 5%

of both the ELG and GLE hypotheses, implying feedback causality between export and economic growth in OECD countries. This finding is consistent with the view in Bhagwati (1988) that higher export leads to more income, which triggers more trade. The country-specific results signal the importance of using various methods to better understand the nature of causality. The different approaches lead to different country-specific results as it is expected.

The results in Table 14 overall throw light on the importance of factor modeling to account for cross-sectional dependence. Using the LA-VAR approach with factor model framework leads the country-specific results to differ substantially. Specifically, the LA-VAR with PANIC and PANICCA approaches compared to the LA-VAR leads to change from causality to non-causality for Finland, Mexico, S. Korea, and Sweden;

Table 14 Summary of the panel causality tests

	SUR	VAR	LA-VAR	LA-VAR (PANIC)	LA-VAR (PANICCA)
Panel results		↔	↔	↔	↔
Country-specific results					
Australia				←	←
Austria		→			
Canada					
Denmark		←			
Finland		←	↔		
France				←	←
Germany					
Italy		←			
Japan					
Mexico	←		←		
Netherlands	→				
New Zealand					
Norway				←	←
Portugal					
S. Korea		↔	→		
Spain					
Sweden		←	←		
Türkiye					
UK	←				
USA				→	→
# Rejection (5%)	3	7	5	4	4

The number (#) of rejection is based on the adjusted *p*-values using the Holm's procedure at 5%
→: One-way causality from export to GDP
←: One-way causality from GDP to export
↔: Two-way causality between export and GDP

and change from non-causality to causality for Australia, France, Norway, and USA. Furthermore, the causality approach based on PANIC and PANICCA procedures indicates that the EU countries (with two exceptions, France and Norway) do not have any causality between export and economic growth while other methods provide mixed results for the direction of causality. This finding can plausibly be explained by some arguments. Hagemejer and Mućk (2019) claim that the continued investment needs of a growing economy are likely to drag down the overall contribution of net exports to growth in a highly export-oriented developing country with fast export growth. The other possibility is about trade barriers. Some studies argue that a rise in trade barriers would have the negative effects on growth (see among others, Head and Mayer 2016; Dhingra et al. 2017). Especially, trade policy regulations in USA and the Brexit

negotiations have affected the trade in the EU (Altomonte et al. 2018). Finally, such a finding can broadly be associated with competitiveness. Turégano and Marschinski (2020) show that the EU´s global share diminished in last two decades due to an overall structural shift in the composition of final demand away from manufacturing goods. They also find that participation losses significantly declined the EU´s share in global manufacturing value chains.

5 Concluding remarks

This paper extends the literature on testing for Granger causality in heterogeneous panels by accounting for cross-sectional dependence. We first estimate unobserved common factors with the PANIC and PANICCA approaches and remove cross-sectional dependence from the data. We then estimate the LA-VAR model for each cross section and use a simple way to construct the panel statistics by combining the p-values of the individual Wald statistics.

We examine the small sample properties of the proposed tests under different data-generating processes with different unit root and co-integration properties for cross-sectional units. The simulations show that although the test statistics show mild size distortions for small samples, they have good size properties with an increasing time dimension. Moreover, the test statistics have high empirical power even in small samples.

The empirical application re-investigates Granger causality between export and economic growth in OECD countries. The findings uncover a crucial role of considering common factors for country-specific inferences and reveal less evidence on causality in the majority of the EU countries.

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Appendix: Simulation evidence on Conjecture 1

This section carries out simulations to support the Conjecture 1, by examining whether the de-factoring procedure effects the asymptotic properties of W_i statistic. To this end,

we investigate how W_i applied to estimated idiosyncratic component behaves in comparison to it applied to true idiosyncratic component. The data is simulated under the null hypothesis, given by $Z_{it}^{cd} = \Lambda_i' F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$, $u_t, \varepsilon_{it}, \Lambda_i \sim iidN(0,1)$ for $T \in \{50, 100, 200, 300, 400\}$ and $N \in \{20, 50, 100\}$. The common and idiosyncratic components are estimated by principal components analysis applied to the first-differences. Monte Carlo simulations are conducted with 50,000 replications to compare mean, variance, and rejection rate (using χ_1^2 critical value at 5%) of W_z and $W_{\widehat{z}}$ statistics, which respectively denote W_i statistic applied to the true (Z_{1t}) and the estimated ($\widehat{Z}_{1t}$) idiosyncratic components.

The results in Table 15 show that $W_{\widehat{z}}$ and W_z statistics appear to have not only the same mean and variance, which are quite close to the mean and variance of the χ^2 distribution with $E(\chi_1^2) = 1$ and $var(\chi_1^2) = 2$, but also the same rejection rate under the null hypothesis. Figure 2 plots the histograms of simulated W_z and $W_{\widehat{z}}$ statistics,

Table 15 Simulation results for Conjecture 1

N	T	W_z			$W_{\widehat{z}}$		
		Mean	Variance	Rejection rate	Mean	Variance	Rejection rate
20	50	1.106	2.607	0.062	1.104	2.605	0.062
	100	1.053	2.295	0.057	1.056	2.309	0.058
	200	1.023	2.126	0.053	1.022	2.118	0.052
	300	1.013	2.083	0.052	1.01	2.075	0.051
	400	1.012	2.071	0.051	1.012	2.084	0.051
50	50	1.124	2.786	0.065	1.119	2.756	0.064
	100	1.059	2.327	0.058	1.060	2.332	0.058
	200	1.019	2.071	0.053	1.019	2.073	0.052
	300	1.015	2.063	0.053	1.015	2.070	0.052
	400	1.013	2.034	0.052	1.012	2.029	0.052
100	50	1.107	2.662	0.063	1.104	2.665	0.063
	100	1.055	2.316	0.058	1.053	2.305	0.057
	200	1.022	2.145	0.052	1.023	2.132	0.053
	300	1.017	2.050	0.052	1.018	2.064	0.052
	400	1.005	2.029	0.051	1.007	2.038	0.051

The mean and variance are based on Monte Carlo simulations with 50,000 replications. The DGP is given by $Z_{it}^{cd} = \Lambda_i' F_t + Z_{it}$, $F_t = \phi F_{t-1} + u_t$, $Z_{it} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} Z_{it-1} + \varepsilon_{it}$ with $\phi = 0.5$ $u_t \sim iidN(0,1)$, $\varepsilon_{it} \sim iidN(0,1)$, and $\Lambda_i \sim N(0,1)$. The rejection rate is based the 5% critical value of χ_1^2 under the null hypothesis. W_z denotes the individual Wald statistic applied to true idiosyncratic components Z_{1t}. $W_{\hat{z}}$ denotes the individual Wald statistic applied to the estimated idiosyncratic components $\hat{Z}_{1t}$. We use the first-differenced data to estimate common and idiosyncratic components by means of principal components analysis

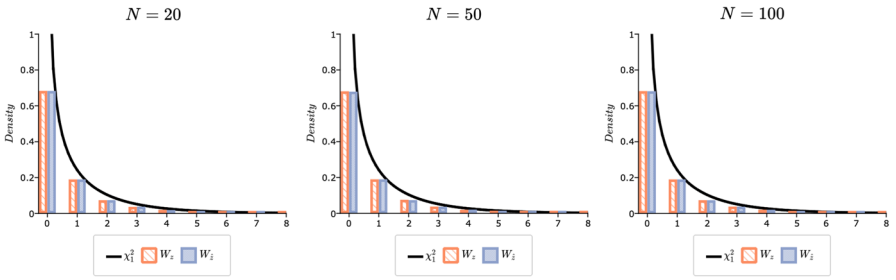


Fig. 2 Histogram of simulated W_z and $W_{\widehat{z}}$ (compared to χ^2_1 density)

along with the density of χ^2 distribution for $N \in \{20, 50, 100\}$ and $T = 1000$. It, furthermore, displays that the distributions of W_z and $W_{\widehat{z}}$ appear identical and well match the χ^2 distribution. Thereby, the de-factoring step appears to have no consequence on the asymptotic properties of W_i statistic, which tends to have an χ^2 distribution with k_i degrees of freedom.

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